EECS 4080 – Supervisor Feedback

Milestone 4: Embedding Vector DB Classifier, LLM-as-a-Judge & Automated Benchmarking (*Reviewed via written submission and GitHub repository)*

Overall Assessment

Milestone 4 is complete and well-executed. Both alternative paradigms are correctly implemented, properly evaluated against the frozen test set, and the automated benchmarking aggregator is clean and functional. Your engineering analysis for each paradigm goes beyond surface-level reporting: the explanation of why KNN fails (semantic overlap in vector space) and why LLM-as-a-Judge fails (instruction-following susceptibility) demonstrates genuine understanding of the architectural trade-offs at play. All numbers in the written report are consistent with the committed code and results files.

Two items require your attention before Milestone 5: the F1 target of ≥ 85% remains unmet at 83.31%, and the OOD / BIPIA evaluation has not yet been started. The BIPIA evaluation is the highest-risk remaining deliverable in the project and must be prioritised immediately.

What Went Well

Paradigm A: Embedding Vector Database (Semantic Search)

- Correct architecture. all-MiniLM-L6-v2 maps prompts into a 384-dimensional vector space; KNN (k=5, cosine similarity) performs classification. Implementation is clean and matches the description in your report exactly.
- Latency correctly measured. Inference is run at batch_size=1 inside a per-prompt loop, matching the real-world sequential traffic methodology you established in Milestone 3. Result (12.96 ms) is plausible and consistent.
- Failure analysis is substantive. You identified the root cause correctly: benign prompts containing complex instructions ("Ignore my previous typo…") sit too close in vector space to jailbreak prompts ("Ignore previous instructions…"), and KNN has no mechanism to weight the decision boundary asymmetrically the way your DistilBERT loss function does.

Paradigm B: Local LLM-as-a-Judge (Zero-Shot Generative AI)

- DoS vulnerability identified and fixed proactively. Discovering that prompt #407 (2,254 tokens) overflowed the context window and implementing both an n_ctx expansion to 4,096 and a left-truncation strategy is a genuine engineering contribution. It was not required by the contract but directly improves reproducibility and robustness.
- System prompt design is appropriate. Constraining the model to output exactly one word ("SAFE" or "ATTACK") with temperature=0.0 and max_tokens=5 is the correct approach for deterministic binary classification from a generative model.
- Failure analysis is the strongest piece of writing in this submission. The observation that LLMs are inherently vulnerable to the very attack type they are classifying is a precise and insightful finding.

Automated Benchmarking Aggregator

- Clean and defensive implementation. The script sweeps results programmatically, handles missing files with a clear warning rather than crashing, and prints a Pandas-formatted leaderboard with automatic contract verdict logic (FPR < 5% pass/fail).
- The master_benchmark_report.json output creates a single source of truth for all four-model metrics, which will be the foundation of your final technical report.

Contract Metrics Summary

Metric	Your Result	Contract Target	Status
DistilBERT F1-Score	83.31%	≥ 85%	Below Target
DistilBERT FPR	4.05%	< 5%	PASSES
Inference Latency	10.9 ms	< 50 ms	PASSES
Baseline Comparison (F1)	+17.26% vs DeBERTa	≥ baseline or ±5%	PASSES
Vector DB FPR	21.92%	Informational	FAILED
LLM-as-a-Judge FPR	15.92%	Informational	FAILED
OOD / BIPIA Evaluation	Not yet started	F1 ≥ 0.70 (Milestone 5)	PENDING

Final Architectural Standings

Model	F1-Score	FPR	Recall	Precision	Latency	Verdict
ProtectAI DeBERTa	66.05%	21.32%	65.30%	66.82%	—	**FAILED**
Custom DistilBERT	83.31%	4.05%	75.80%	92.48%	10.9 ms	**PASSED**
Vector DB (KNN)	79.84%	21.92%	88.58%	72.66%	12.96 ms	**FAILED**
LLM-as-a-Judge (Phi-3)	68.44%	15.92%	64.61%	72.75%	539.1 ms	**FAILED**

Gaps and Required Actions

1. F1 Score — Still Below Contract Target

Your contract specifies F1 ≥ 85%. Your current test set result is **83.31%**, a gap of 1.69 percentage points. You have acknowledged this shortfall clearly in both your Milestone 3 revision and Milestone 4 report and attributed it correctly to the jailbreak-heavy dataset distribution. You have committed to addressing this in Milestone 5 via hyperparameter tuning.

For Milestone 5, please document the specific tuning attempts you make (e.g., learning rate, epoch count, w_benign coefficient adjustments) and report whether the gap was closed. Even if 85% is not reached, a clear record of what was tried and why the gap persists is acceptable for the final report — the contract acknowledges that dataset difficulty is a mitigating factor.

2. OOD / BIPIA Evaluation — Must Begin Now

This is the highest-risk remaining item in the project.

The contract explicitly requires testing against the BIPIA dataset (indirect injections unseen during training), with a target OOD F1 ≥ 0.70. This evaluation has not been mentioned in any submission prior to the Midterm Plan Forward section and no sourcing or protocol design is visible in the repository.

This is a non-trivial deliverable. It requires:

- Sourcing the BIPIA dataset (it is publicly available on Hugging Face).
- Designing an OOD evaluation protocol, the dataset format must be aligned to your existing pipeline (text/label columns, the same frozen test set discipline).
- Running inference with your trained DistilBERT model on BIPIA prompts it has never seen.
- Reporting the OOD F1 and comparing it against the ≥ 0.70 contract target.

Milestone 5 is due **July 19**. Please begin sourcing BIPIA and designing the evaluation protocol immediately, do not leave this to the final week.

3. Minor: KNN vs. Contract-Specified Downstream Classifier

The contract specified the Alternative ML approach as an embedding-based classifier using "Random Forest or XGBoost" as the downstream component. You implemented KNN instead. KNN is a reasonable and well-justified choice for cosine similarity search, and your engineering rationale is sound. However, please add one sentence to your final report noting this deviation and briefly justifying the choice. This is a minor documentation gap, not a substantive concern.

Looking Ahead — Milestone 5 Checklist

- **F1 Target:** Attempt F1 ≥ 85% via hyperparameter tuning; document all attempts.
- **BIPIA (Priority):** Source BIPIA dataset and design OOD evaluation protocol.
- **OOD Evaluation:** Run OOD inference with trained DistilBERT; report F1 against ≥ 0.70 target.
- **Final Report Draft:** Draft final technical report including background, system design, benchmarking data, and trade-off analysis.
- **KNN Note:** One sentence noting KNN deviation from contract spec and justification.